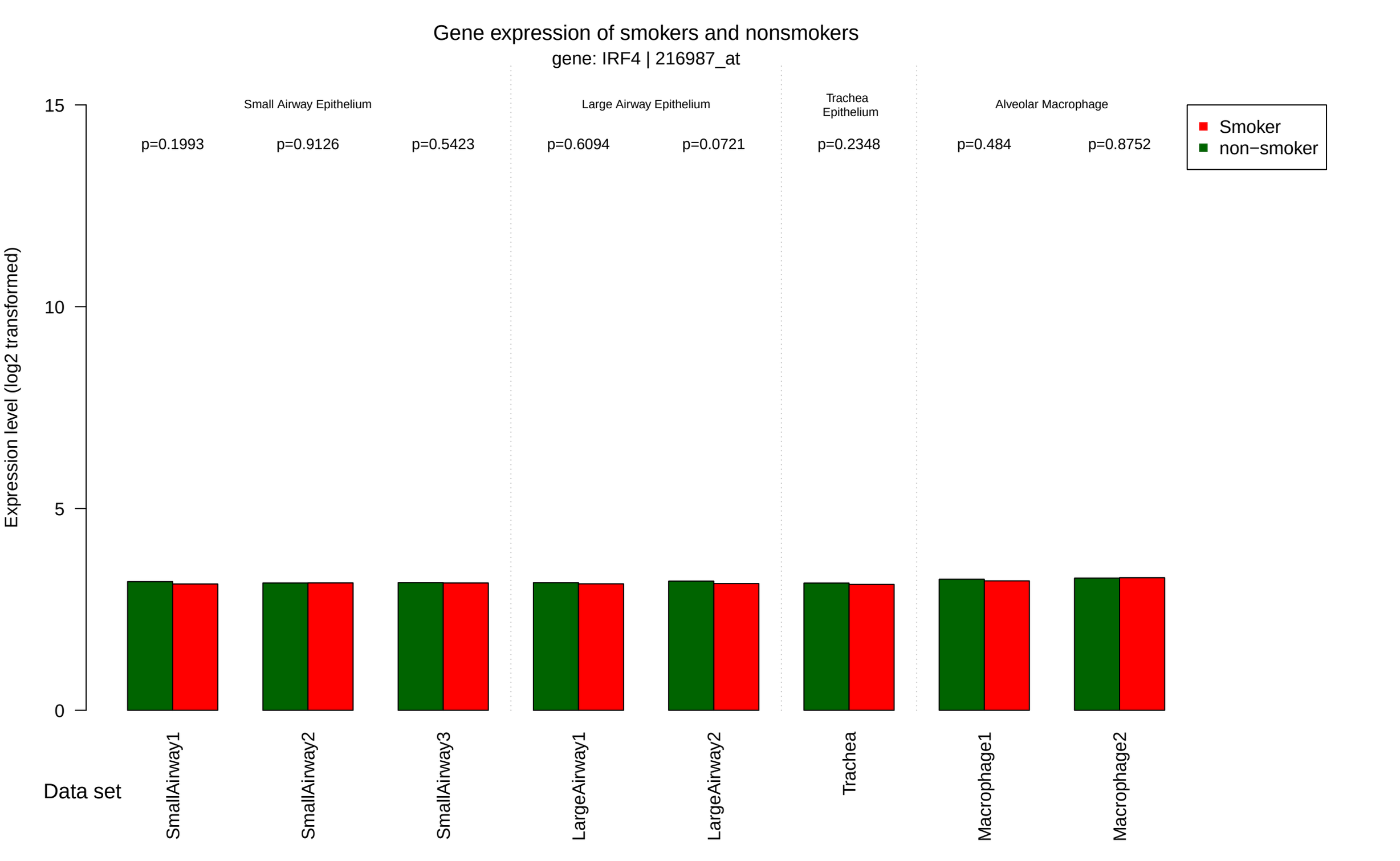
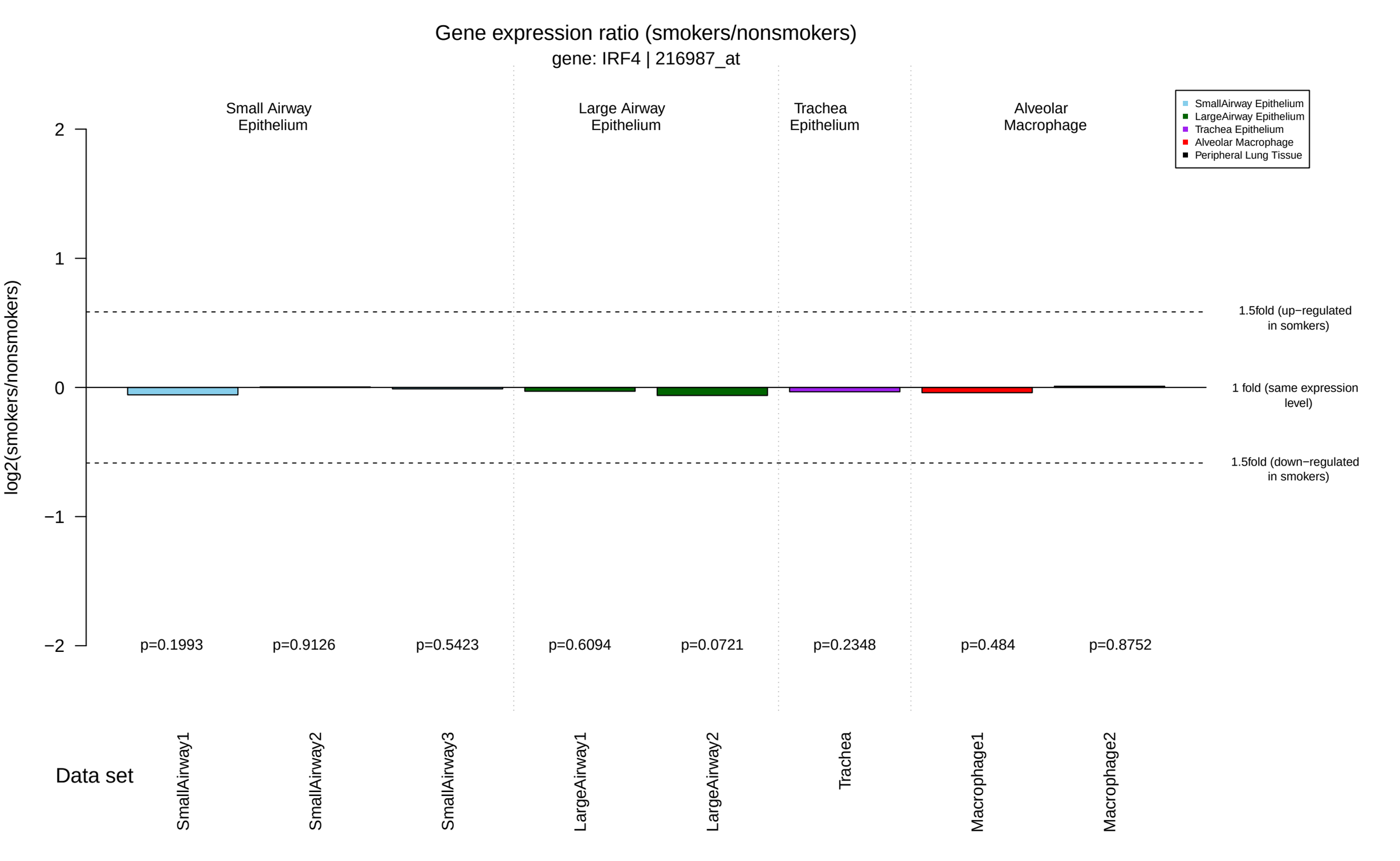
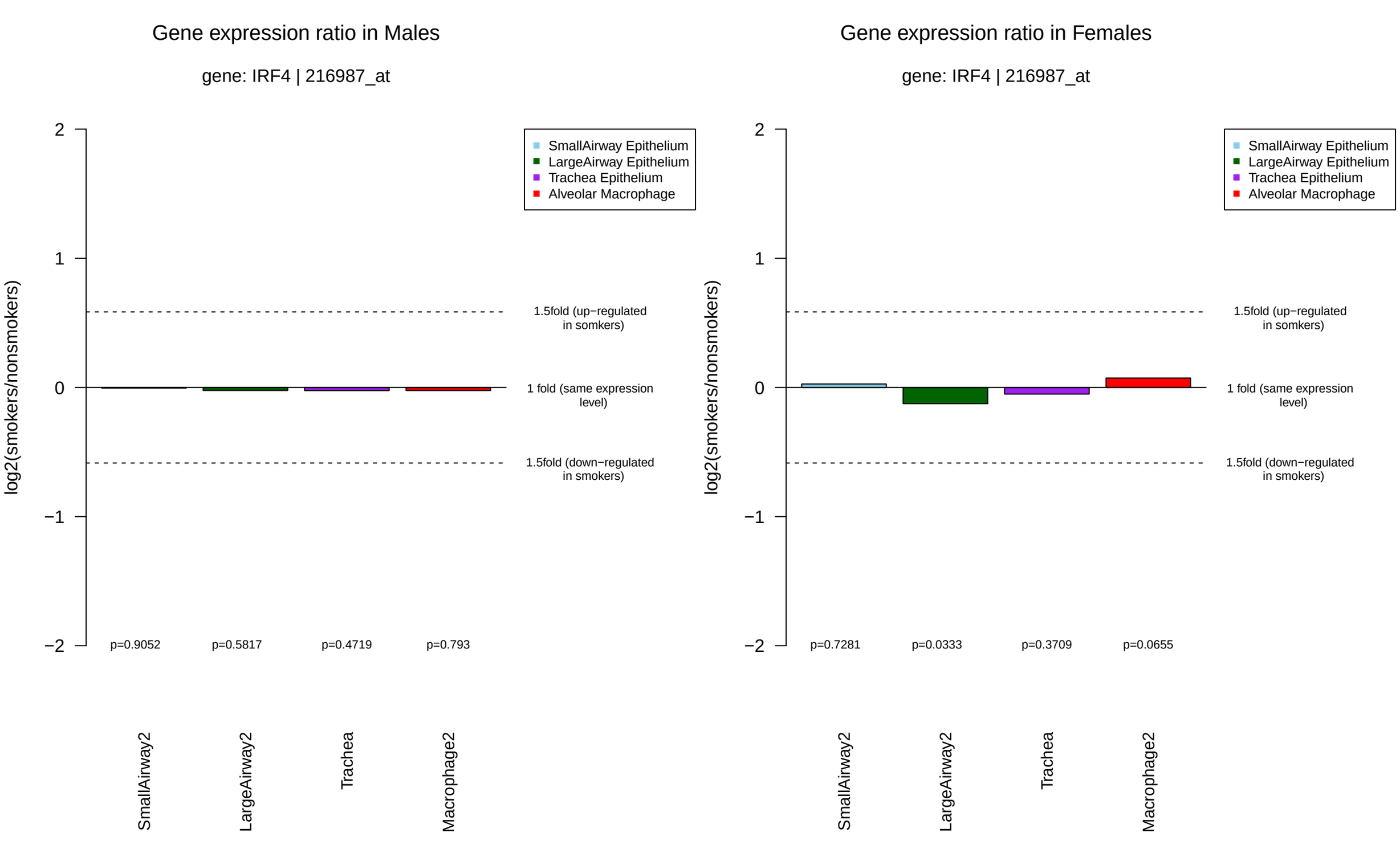
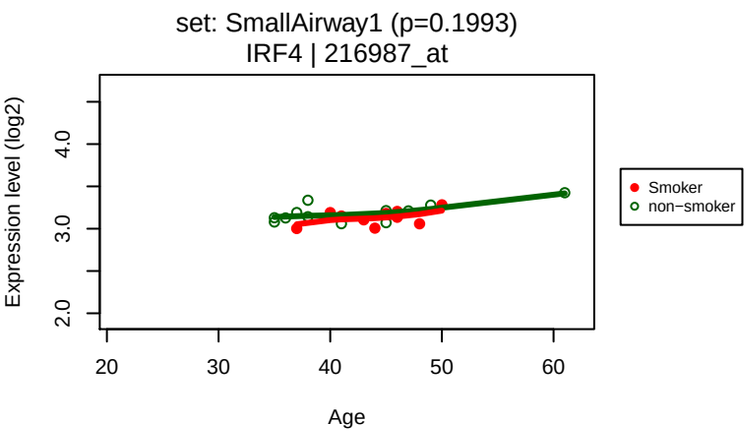
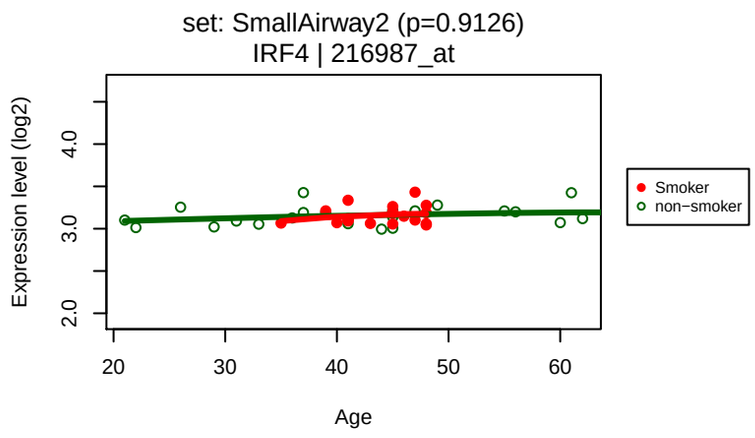
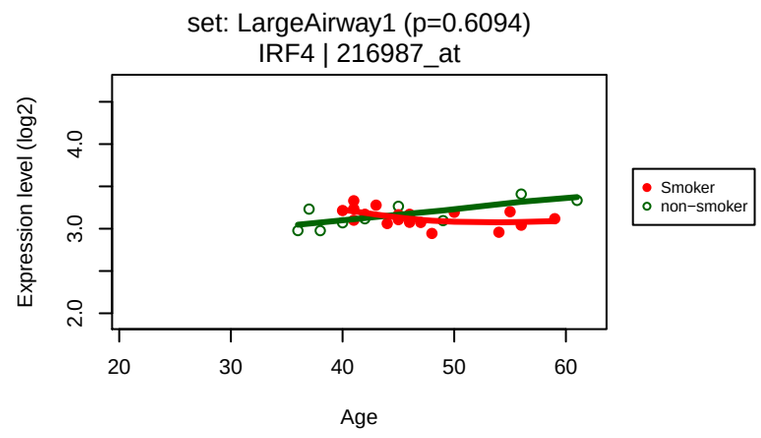

Gene expression of smokers and nonsmokers
gene: IRF4 | 216987_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1993
p=0.9126
p=0.5423
p=0.6094
p=0.0721
p=0.2348
p=0.484
p=0.8752
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

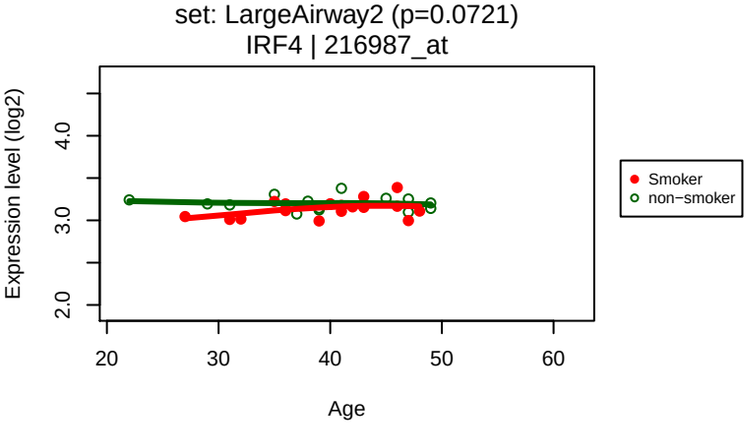

Gene expression ratio (smokers/nonsmokers)
gene: IRF4 | 216987_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1993
p=0.9126
p=0.5423
p=0.6094
p=0.0721
p=0.2348
p=0.484
p=0.8752
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

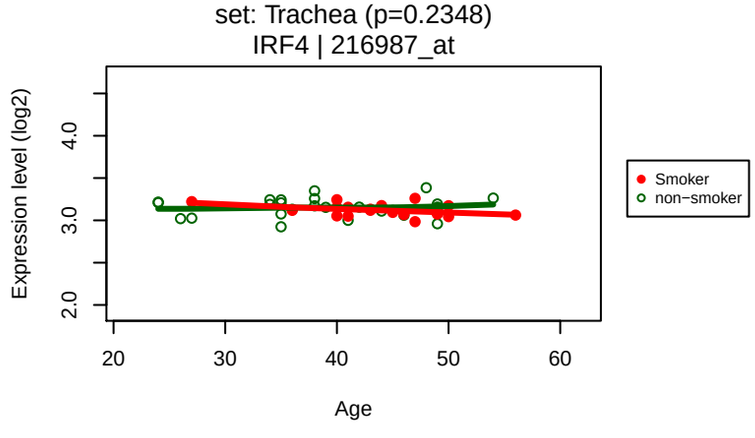

Gene expression ratio in Males
gene: IRF4 | 216987_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9052
p=0.5817
p=0.4719
p=0.793
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRF4 | 216987_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7281
p=0.0333
p=0.3709
p=0.0655
SmallAirway2
LargeAirway2
Trachea
Macrophage2

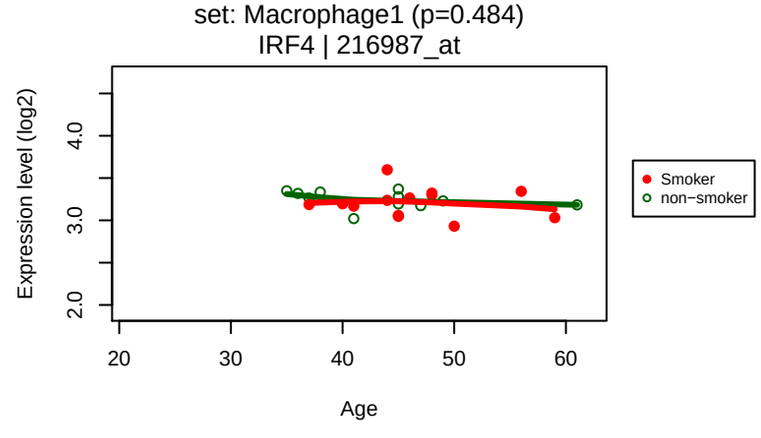

set: SmallAirway1 (p=0.1993)
IRF4 | 216987_at
Expression level (log2)
Age
Smoker
non−smoker

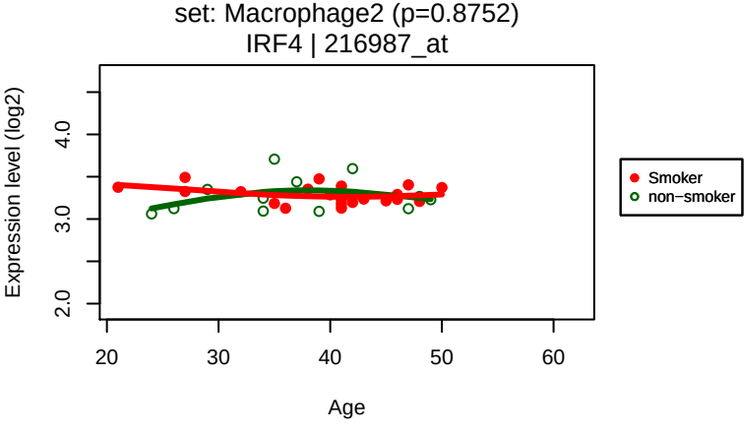

set: SmallAirway2 (p=0.9126)
IRF4 | 216987_at
Expression level (log2)
Age
Smoker
non−smoker

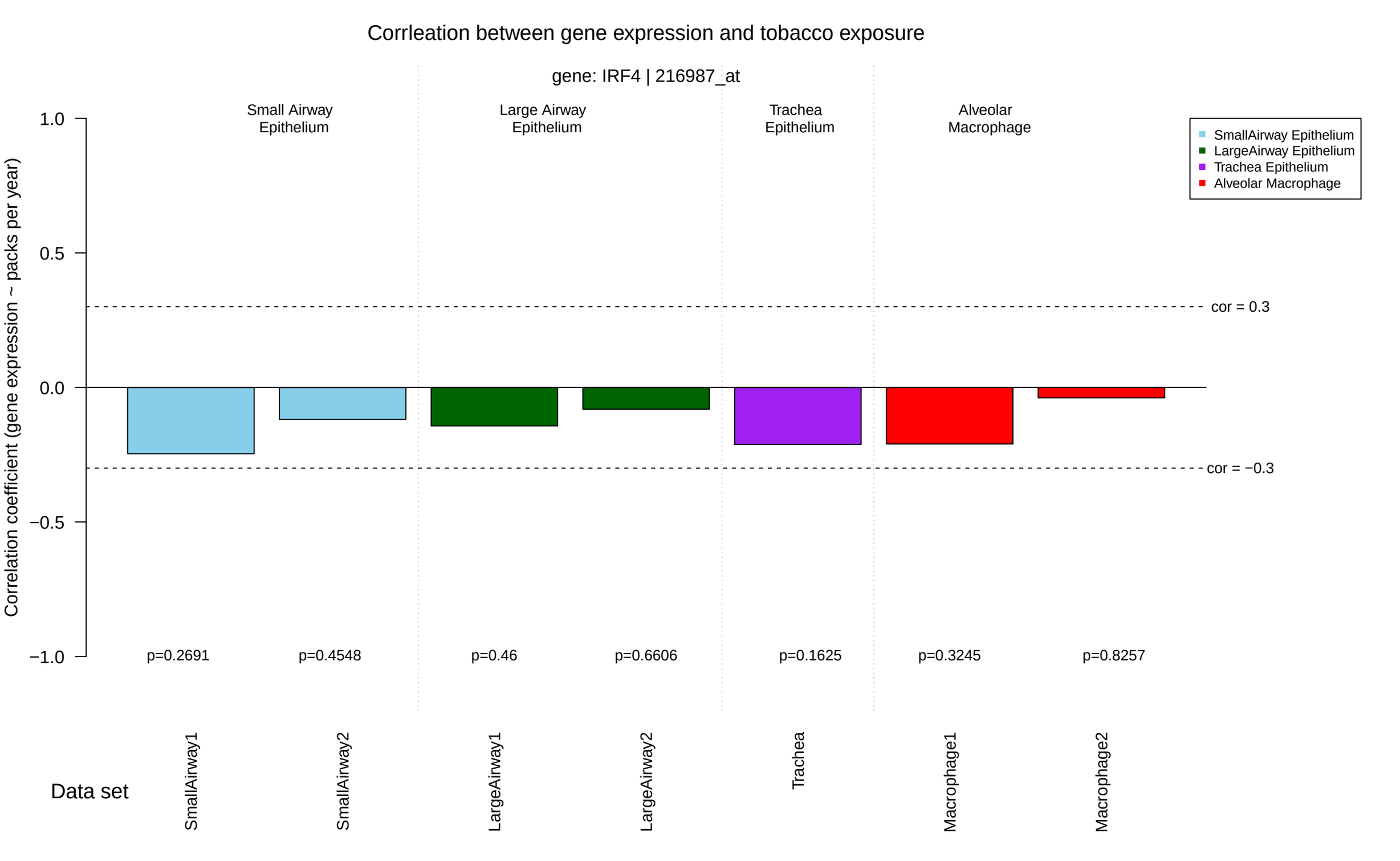

set: LargeAirway1 (p=0.6094)
IRF4 | 216987_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0721)
IRF4 | 216987_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2348)
IRF4 | 216987_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.484)
IRF4 | 216987_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8752)
IRF4 | 216987_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRF4 | 216987_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2691
p=0.4548
p=0.46
p=0.6606
p=0.1625
p=0.3245
p=0.8257
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2